12 May 1971
SUPERSEDING
MIL-E-1/296B
17 April 1957

MILITARY SPECIFICATION SHEET

ELECTRON TUBE REFERENCE CAVITY

TYPES 1Q22, 1Q23, 1Q24, 5846, 6040, 6041

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Reference cavity stabilizer, X-Band, Waveguide mounted.

ABSOLUTE RATINGS:

	Resonant Frequency (MHz)						
	1Q22	1Q23	1Q24	5846	6040	6041	Altitude
Minimum:	9249.5	9279.5	9309.5	9279.5	9307.5	9311.5	--------
Maximum:	9250.5	9280.5	9310.5	9280.5	9308.5	9312.5	50,000 ft.

PHYSICAL CHARACTERISTICS:

Dimensions:	See outline drawing
Weight:	8 ounces (approximately)
Mounting Position:	Any

TEST CONDITIONS: See note 1

GENERAL:

Qualification: Required

FSC 5960

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1					
----	Insertion loss	At resonant frequencies	Li	4.0	6.0	db
----	Loaded Q		QL	1900	2400	---
----	Temperature cycle (1)	T = 40° to 100°C t = 30 minutes See note 4	1Q22F_o 1Q23F_o 1Q24F_o 5846F_o 6040F_o 6041F_o	9249.7 9279.7 9309.7 9279.7 9307.7 9311.7	9250.3 9280.3 9310.3 9280.3 9308.3 9312.3	MHz MHz MHz MHz MHz MHz
----	RF Glow test	See note 6	----	---	---	---
	Quality conformance inspection, part 2					
1031	Vibration	10 G, See note 3	ΔF_o	---	.15	MHz
1042B	Shock	50 G, 6 ms half-sine	ΔF_o	---	.15	MHz
--	Temperature cycle (2)	T = 40° to 100°C t = 12 hours See note 4	1Q22F_o 1Q23F_o 1Q24F_o 5846F_o 6040F_o 6041F_o	9249.7 9279.7 9309.7 9279.7 9307.7 9311.7	9250.3 9280.3 9310.3 9280.3 9308.3 9312.3	MHz MHz MHz MHz MHz MHz
----	Temperature compensation (1)	Room temperature to 100°C, See note 5	ΔF_o	---	0.3	MHz
----	Off resonance insertion loss	F = 9300 ± 100 MHz F = 3350 ± 100 MHz	Li Li	30 30	--- ---	db db
4003	Pressurizing	45 Psi absolute See note 9	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3					
----	Life test	Group B t = 30 minutes T = 55° to 100°C See note 7	Cycles	25	---	---
----	Life test end points					
----	RF Glow test	See note 6	---	---	----	---
----	Resonant frequency		$1Q22F_o$	9249.5	9250.5	MHz
			$1Q23F_o$	9279.5	9280.5	MHz
			$1Q24F_o$	9309.5	9310.5	MHz
			$5846F_o$	9279.6	9280.4	MHz
			$6040F_o$	9307.5	9308.5	MHz
			$6041F_o$	9311.5	9312.5	MHz
	Periodic check tests	See note 2				
----	Temperature cycle (3)	T = 55° to 100°C t = 30 minutes See note 4	$1Q22F_o$	9249.7	9250.3	MHz
			$1Q23F_o$	9279.7	9280.3	MHz
			$1Q24F_o$	9309.7	9310.3	MHz
			$5846F_o$	9279.7	9280.3	MHz
			$6040F_o$	9307.7	9308.3	MHz
			$6041F_o$	9311.7	9312.3	MHz
----	Temperature compensation (2)	Room temperature to 0°C	ΔF_o	---	0.3	MHz
		Room temperature to -25°C	ΔF_o	---	0.7	MHz
		Room temperature to -40°C	ΔF_o	---	1.0	MHz
		Room temperature to -55°C See notes 5 and 8	ΔF_o	---	1.2	MHz
----	Atmospheric pressure test	P = Atmospheric to 45 psi	ΔF_o	---	0.15	MHz
		P = Atmospheric to 5 in. Hg	ΔF_o	---	0.15	MHz

NOTES:

1. Electrical measurements are to be made with a standard frequency source having an absolute accuracy of the order of 1 part in 500,000. The cavities shall be tested mounted between UG-40/U chokes in RG-52/U waveguide and with a voltage standing wave ratio less than 1.05 due to resistive match looking into each of the chokes between which the cavity is mounted.

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NOTES: -Continued

2. If these tests have been previously performed by the manufacturer, and if the material, manufacturing process, and test requirements are the same as when the tests were performed, the subject tests need not be performed.

3. The difference in resonant frequency before and after vibration shall not exceed the limits specified.

4. The cavity shall be subjected to three complete cycles between the specified temperature extremes. The cavity may be allowed to come to equilibrium at room temperature in passing from one extreme to the other. The cavity shall be maintained at the extreme temperatures for the specified time during each cycle. When the tube has come to equilibrium at room temperature at the conclusion of the last two half cycles the resonant frequency shall be measured and shall be within the specified limits.

5. After temperature cycling described in Note 4, the cavity shall be brought to equilibrium at the specified extreme temperature for 30 minutes and then returned to equilibrium at room temperature. Then it shall be brought to the specified extreme temperature again and allowed to return to room temperature. Resonant frequency measurements shall be made at approximately 20°C temperature intervals while returning to room temperature. At no temperature shall the resonant frequency differ from the value at room temperature by more than the specified amount.

6. The tube shall show a typical diffuse low-pressure glow discharge when placed in an rf field.

7. The cavity may be allowed to come to equilibrium at room temperature in passing from one extreme to the other. The cavity shall be maintained at the extreme temperature for the specified time during each cycle. When the tube has come to equilibrium at room temperature at the conclusion of the last two half cycles, the resonant frequency shall be measured and shall be within the specified limits.

8. ΔF_o at -55°C may be obtained by extrapolation of data obtained at higher temperatures.

9. This test is not applicable to type 5346.

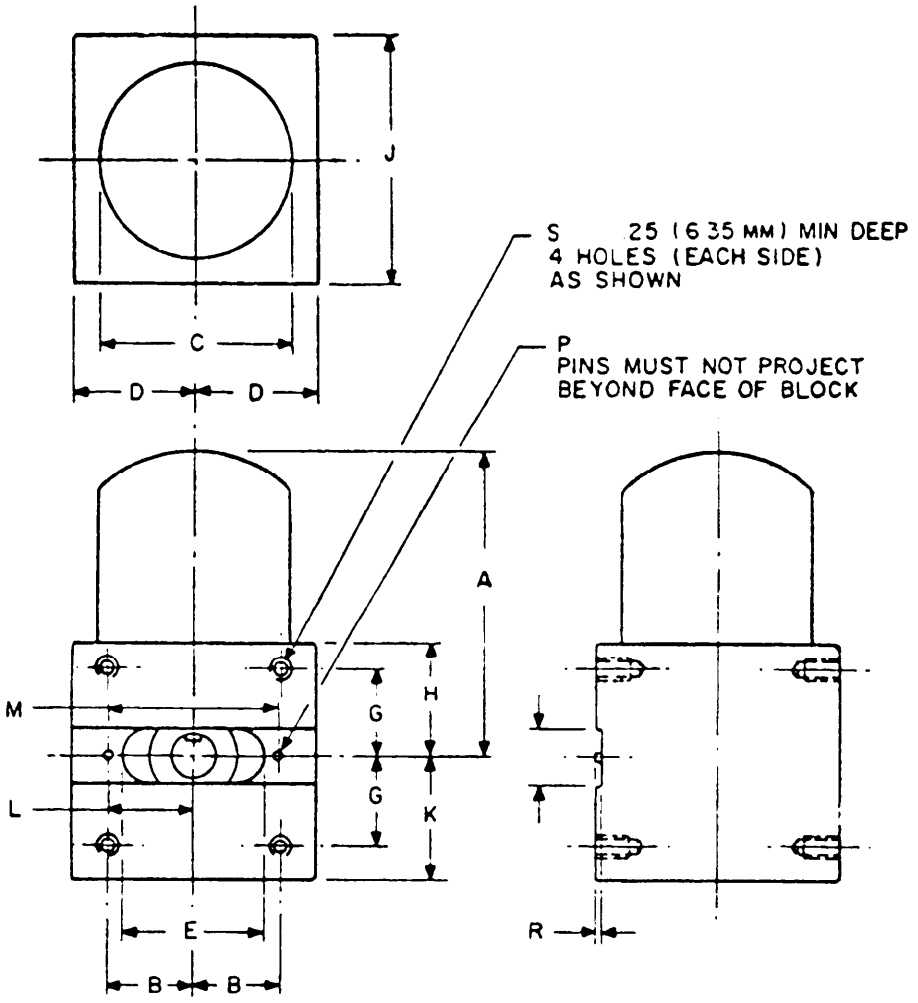
Custodians:
Air Force - 80
Army - EL
Navy - EC

Preparing activity:
Air Force - 80

Review activities:
Air Force - 11,
Army - MU
Navy - SH

(Project 5960-2425-55)

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DIMENSIONS									
LTR	INCHES		MILLIMETERS		LTR	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX		MIN	MAX	MIN	MAX
QUALITY CONFORMANCE INSPECTION, PART 1					D	.859	.891	21.82	22.63
J	1.744	1.756	44.30	44.60	E	1.000 REF		25.4 REF	
M	1.218 NOM		30.94 NOM						
• P	.061	.065	1.55	1.65	G	.638	.642	16.21	16.31
• Q	.400	.402	10.16	10.21	H	–	.828	–	21.03
					K	–	.875	–	22.23
QUALITY CONFORMANCE INSPECTION, PART 2					• L	.609 NOM		15.47 NOM	
A	–	2.187	–	55.55	• N	.020	.030	.51	.76
B	.608	.612	15.44	15.54	• R	.028	.030	.71	.76
C	–	1.375	34.	34.93	S	8-32 UNC-2B THD			

• THESE DIMENSIONS APPLY ONLY TO 5846

FIGURE 1. Outline drawing of tube types 1Q22, 1Q23, 1Q24, 5846, 6040 and 6041

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